

Generative AI and Large Language Model Prompting for Hazard Forecasting

Angus Dagg, Spencer Gable, Alberto Ardid, Daniel Nielsson, David Dempsey

*ENCN 493 2025 – Research Project
Department of Civil and Natural Resources Engineering
University of Canterbury, Christchurch, New Zealand*

ABSTRACT

Creation of emergency alert messages relies on an understanding of human behavioural responses, but human experiments are time-consuming, costly, and potentially unsafe for participants. This study investigates the use of large language models (LLMs) as simulators of human behaviour in these experiments. Using a Python pipeline, LLMs were prompted with persona agents and a variety of emergency messages, replicating the methodology of Carlson and Barbour (2023). A Likert scale was used to record ratings of safety and message efficacy, with the effects of prompt structure and temperature settings examined. Results showed that persona prompting and larger models had the greatest effect on shifting responses. Beyond this study, future research could compare models of varying size and architecture to examine LLM progression over time and test this pipeline in instinct-driven contexts such as fire evacuation or earthquake response. While the use of LLMs as proxies for human participants shows promise, further fine-tuning on prompting techniques will be required.

Keywords: Large language models (LLMs), emergency alert messaging, persona prompting, behavioural simulation

1 INTRODUCTION

Emergency alert messages are essential for public safety, but predicting human behavioural responses is challenging. Traditional surveying methods suffer from biases and require significant methodological expertise (Kennedy, Jensen & Jensen, 2022). With the increasing capabilities of large language models (LLMs), there is a promising opportunity to integrate them into behavioural response studies. Recent research points towards their emerging potential in this field, but it remains unclear how accurately they can imitate alert message responses (Maier et al., 2025).

This study explored whether LLMs could simulate realistic human responses to mobile alert messages, offering a verification means or alternative method to human surveys. Specifically, it examined how LLM temperature settings (which control output randomness) and various persona prompting techniques influenced responses. It also identified effective prompt engineering strategies for guiding responses in a desired direction. To do this, a coded pipeline was developed to prompt LLMs with personas and emergency alert messaging, recreating the methodology of Carlson and Barbour (2023).

If validated, LLM-generated responses could assist in early-stage human research and enhance survey reliability by serving as a verification method. Consequently, this study addresses three research questions:

RQ1: What effect do temperature values have on emergency alert message responses?

RQ2: How do different persona prompting techniques influence the variability and realism of simulated responses?

RQ3: To what extent can large language models replicate real-world emergency alert message survey results?

2 LITERATURE REVIEW

2.1 Challenges in studying human behavioural responses

Traditional behavioural intent studies are subject to the shortcomings of human psychology. For instance, participants often display the self-presentation theory, whereby subjects attempt to answer with what they think the examiner wants to hear, as opposed to honestly (Goffman, 1959). Furthermore, participants are also influenced by the theory of reasoned action. This suggests that attitude does not relate to behaviour unless the individual has previously encountered the situation. The theory implies that you do not know how to act until you're forced to act. Regarding behavioural studies, participants use understanding of their attitudes to infer how they would behave as opposed to their actual behavioural intent (Fishbein & Ajzen, 1974). Conventional LLMs suffer from similar shortcomings, with reinforcement learning (a post-training process where models are rewarded for producing preferred responses) promoting social desirability and imagined scenarios.

Beyond psychological biases, there is the risk of bias from the survey methodology. Traditional online surveys can suffer from undercoverage bias (Bethlehem, 2010), whereby results are only representative of a subpopulation with internet access. This is similar to non-response bias, which occurs when the people who don't complete a survey are systematically different from those who do (Sandelin, 2022). LLM survey responses would not face the same sample size limitations traditional surveys struggle with since they are infinitely scalable and can, in principle, represent broader demographics.

Questionnaire length is another key constraint to human survey responses, with Sandelin (2022) reporting a 2.7% increase in response rate when going from a twelve-page to a nine-page survey. LLM survey responses could improve on this, not experiencing questionnaire fatigue like humans do. Extreme instances may lower LLM response quality, as large prompts can make them more susceptible to errors (Du et al., 2025).

Together, these issues highlight the limitations of traditional human behavioural response surveys and suggest that LLMs may mitigate, though not eliminate, their effects.

2.2 LLMs as behavioural simulators

Recent research points towards the emerging abilities of LLMs to display human-like behaviour (Binz et al, 2025). Although morality was once thought difficult for LLMs to capture, Dillon et al., (2023) found GPT-3.5's judgments closely aligned with human responses ($r = 0.95$) across hundreds of scenarios. Variation from human decision-making appears to occur in cases of competing intuitions, where LLMs must prioritize one moral judgment over another (Dillon et al, 2023). This is relevant to emergency messaging, where moral reasoning might, for example, balance public safety and individual freedom.

A key point of difference may arise from the system prompts used in LLMs, where they are introduced with upstream traits such as high agreeableness and low neuroticism to fulfil their roles as AI assistants (Molchanova et al, 2025). These traits are often accentuated in post training, where reinforcement learning further embeds socially desirable behaviours. Given that these traits are not shared by all humans, more accurate LLM persona databases may need to selectively override system prompts on a local model or adjust post-training procedures to better reflect diverse human behaviours.

Even with their recent success, LLMs do not yet perfectly simulate human responses. Zhang, Zu & Alvero (2024) found that while LLM survey responses were longer and more lexically diverse than human ones, this did not translate into richer content. This highlights the absence of lived emotional experience in LLMs, which may limit their ability to capture nuance in contexts, such as emergency alerts, where urgency and fear strongly shape compliance. While LLMs draw only on written text, images, and audio, humans base decisions on embodied and emotional experience. While it is possible to persona prompt the AI to respond as if it had experienced a particular emergency event, it might not be able to capture the sentient emotions a human would. LLM responses were also generally more positive and analytical, whereas human responses used more negative and emotionally charged language (Zhang, Zu & Alvero, 2024). This lends itself to persona prompting the LLM to act more pessimistically than it would naturally, a theory tested in this report.

Supporting this, Bisbee et al., (2024) benchmarked LLM responses against American National Election Studies data, finding that although overall trends were captured, important nuances in polarization, subgroup representation, and variance were distorted. The observed differences in stylistic tone and variance reaffirm the importance of calibrating LLM prompting and temperature to better match human survey results.

2.3 Benchmarking against human survey data (Carlson & Barbour)

Carlson and Barbour (2023) was selected to validate the usage of LLM responses in place of human subjects. The goals of their study were to investigate how different message content, style, and structure influence the effectiveness and participants' intended compliance with mobile phone warning messages. The study was conducted on 299 U.S. residents and tested twelve different types of warning messages. Each message was evaluated using a five-point Likert scale, a survey tool that asks participants to rate their agreement with statements in a numerical manner. Ratings were collected for message efficacy (ME) and safety efficacy (SE). **Message efficacy measures a participant's intent to take protective action in response to a warning, while safety efficacy reflects their confidence in making safe decisions after receiving the message.**

Participants were provided by SurveyMonkey and were said to represent the general U.S. population. These criteria introduce the risk of undercoverage bias, with non-mobile phone users and people under the age of eighteen underrepresented. Additionally, because the survey was accessed online, it excluded individuals without internet access. The average age of participants was 36, with a female-dominated population (63%). Participants lived in a range of locations, with varying degrees of education.

Results suggested that prescriptive message types were more effective than notification, and messages that offered choice to the user resulted in higher ME and SE ratings. Replication of the general results will help validate the usage of LLM survey responses.

Due to its use as a validation instrument, the paper's limitations need to be considered. Specifically, the survey only captured 299 results and had an overrepresentation of female and college-educated participants. These caveats should be considered when interpreting the results of any LLM validation.

2.4 Techniques for improving realism in LLM outputs

One promising approach for enhancing the realism of LLM-generated responses is persona prompting, which involves manipulating the LLM to act as a specific character. Recent studies have shown that persona prompting can increase lexical diversity, particularly in larger models, though diminishing returns occur with excessive context (Kambhata et al., 2025). Moreover, persona prompting has been shown to improve Theory-of-Mind reasoning, with LLMs assessing the persona's access to information, answerability, and belief state before formulating a response (Tan et al., 2024). In the context of replicating survey information, this allows for more accurate imitation of the participant's background knowledge and attitudes based on sociodemographic characteristics.

The effectiveness of persona prompting varies depending on the output type. Molchanova et al (2025), found that conveying personas through open-ended text generation responses alone was ineffective, while portraying personality through questionnaire responses had relative success. This characteristic makes the numerical rating system of Carlton and Barbour's experiment favourable when evaluating the influence of personas on results.

Single-shot prompting for LLMs provides the model with an example of the desired answer structure. For instance, the

prompt may request that the LLM explicitly respond with numeric values. This is effective in persona prompting as human-created persona can be provided as a template for the LLM to refer to (Schuller et al, 2024). When analysing larger problems, LLMs can struggle to effectively address the entire prompt. Incremental prompting is a method to avoid this, where a complex prompt is broken into simpler prompts, allowing the model to use a step-by-step analysis and promoting the creation of more detailed personas (Schuller et al, 2024).

Chain-of-Thought (CoT) reasoning is the process models take to explicitly lay out the intermediate steps required in reasoning before coming to a conclusive answer. It stimulates logical, step-by-step thinking and allows for the LLM to make errors, back-track, and explore tangents in the process of solving a problem; much the same as a human would (Guo et al., 2025).

Finally, temperature is an important parameter in LLMs that controls the creativity of a model’s response. Token selection in an LLM is determined on a probabilistic system, which assigns higher probabilities to the more context-relevant tokens. A higher temperature will provide higher weightings to the less probabilistic tokens, enhancing the creativity of responses provided by the LLM. A temperature of more than two creates incoherent text, whilst a temperature of zero always selects the most probable token. A suitable range for temperature is 0.1-1.9 (Li et al, 2025). This was hypothesised to be a relevant factor in recreating survey response variance.

2.5 Ethical and practical considerations

AI-generated qualitative data possesses biases from training, being largely based on Western and middle to high socioeconomic subjects (Marshall et al, 2024). This could reduce the accuracy of LLMs in responses for populations not within that demographic. While this is not relevant for mobile emergency alert messaging, it could create bias in other message types.

Given their exposure to far more linguistic data than any human encounters, LLMs are prone to overfitting superficial statistical patterns and treating them as psychological phenomena (Lin, 2025). To validate their results, LLM responses should replicate the variance and mean statistics of human responses. This will ensure that psychological phenomena are represented, if not explicitly understood.

Beyond these methodological issues, there are broader ethical concerns. Unlike human participants, LLMs do not provide informed consent. This allows researchers to generate responses without the ethical requirements associated with human participants. This undoubtedly speeds up the testing process; however, it holds the engineer liable to any adverse consequences stemming from the experiment. This reaffirms the importance of correct validation against human results.

2.6 Summary and research gap

As this literature review suggests, there is potential for LLMs to supplement traditional methods in behavioural response simulation. Current methodologies introduce cognitive biases and often misrepresent true behavioural instinct. LLMs could bypass these fallacies and potentially offer a solution to represent a wide range of participants. Extensive validation is required to justify their usage, and current literature suggests that LLMs struggle to capture the

finer nuances and stylistic tones of a human response. Techniques such as persona prompting, prompt engineering, and temperature adjustment may improve realism, but their effectiveness in replicating human survey responses remains underexplored.

3 METHOD

This study used a Python pipeline to test the responses of LLM models to emergency alerts. The pipeline ran three main experiments, attempting to answer this paper’s research questions. Initial testing was done on the Llama 3:2.1b LLM, which has an intelligent index score of nine when benchmarked against other models. Comparatively, today’s premier model, GPT 5, scores a 68 in the same index (Artificial Analysis, 2025). Upon confirming that the pipeline worked as desired, it then examined the responses of larger LLMs. The results of this paper were generated using the Gemma3_27b model, with an intelligence index score of 22.

To draw any conclusive results, robust validation against human responses was required. Carlson and Barbour’s (2023) research was used for comparison, providing the messaging context to input into the pipeline. Furthermore, the paper’s analysis of variance (ANOVA) results were used as target values for successful LLM replication.

3.1 Temperature

To answer RQ1 and test the influence of temperature, a sweep of different temperature values was required. The initial range of values was 0.1-1.9; however, the llama 3:2.1b model generated incoherent text with a temperature greater than 1.8. To establish a reliable baseline, the temperature range was changed to 0-1.7, with increments of 0.1. Higher temperature values were tested on smarter models, but generated incoherent commentary and Likert ratings outside of the 1-5 scale. For other tests carried out in the pipeline, 1.0 was used as the temperature because it is the default setting for most LLMs. To isolate the effects of temperature, the experiment for RQ1 used no persona prompts.

3.2 Persona creation

When testing the effects of persona characteristics (RQ2), one-dimensional personas, emphasizing one dominant character trait, were created. Five traits were tested against a baseline with no persona prompting as well as its opposing trait (Table 1).

Table 1: Persona trait pairs used in the pipeline.

Persona 1	Persona 2
Optimist	Pessimist
People-Pleaser	Total Sceptic
Calm-Responder	High-Anxiety
Internal Locus of Control	External Locus of Control
High Agreeableness	High Neuroticism

An internal locus of control refers to the belief that one controls their own outcomes, while an external locus of control believes external factors control one’s future. The LLM would receive a persona prompt reflecting specific character traits, such as: “*You are someone who naturally wants to make others happy and give encouragement*” (People-Pleaser) or “*You are highly sceptical of authority and external advice*” (Total Sceptic).

To answer RQ3, a mock population was created, mimicking Carlson and Barbour’s survey statistics. The participant characteristics were then presented to the LLM by means of a persona prompt: “*You are a {age}-year-old {gender} who lives in a {community} community and has {education} education*”.

3.3 Alert messages participants receive

All experiments provided the LLM with one of twelve possible warning messages. These messages were tailored to the contexts of flooded roads or contaminated household water and were structured as prescriptive (providing safety information) or notification (providing a URL for further information). Stylistically, the messages were either instrumental-only, empathetic, or provided a choice for the participant. The instrumental-only message was a baseline that prescribed action or notice about the emergency. The empathetic message aimed to increase message and safety efficacy by acknowledging the irregularity of the emergency, while choice aimed to do so by empowering the participants. The three message types were each assessed under both prescription and notification messages (Table 2).

Table 2: Warning messages for flooded roads, with different construct and message types (Carlson and Barbour, 2023).

Construct	Instrumental-only	Empathy	Choice
Prescription	Some roads are blocked by flooding. Avoid unnecessary travel. Do not attempt to drive through standing water.	Some roads are blocked by flooding. Avoid unnecessary travel. Do not attempt to drive through standing water. Changing your plans might be difficult, but it’s better to wait for the water to go down than to get stranded out on the road.	Some roads are blocked by flooding. Avoid unnecessary travel. Do not attempt to drive through standing water. You are the best judge of whether you really need to travel today ... and which route you should take.
Notification	Some roads are blocked by flooding. Before traveling, visit yourtown.gov/road conditions to see flood maps.	Some roads are blocked by flooding. Before traveling, visit yourtown.gov/road conditions to see flood maps. Changing your plans might be difficult, but it’s better to wait for the water to go down than to get stranded out on the road.	Some roads are blocked by flooding. Before traveling, visit yourtown.gov/road conditions to see flood maps. You are the best judge of whether you really need to travel today ... and which route you should take.

3.4 Response questionnaire

Participants answered eight questions, evaluating message efficacy (ME) and safety efficacy (SE) using a Likert scale of one (strongly disagree) to five (strongly agree) (Table 3). Item three was reverse-coded for analysis purposes, meaning a higher value represented lower safety efficacy. This encouraged participants to think more about answers and prevented them from agreeing with every message.

Table 3: Survey items measuring Safety Efficacy (SE) and Message Efficacy (ME) (Carlson and Barbour, 2023).

Safety Efficacy	Message Efficacy
If I got this message, I would do what the message says I should do to stay safe.	I feel confident that I could make good decisions about keeping myself safe in a situation like this.
This message would catch my attention.	This message reminded me that I can make good choices about my safety.
This message would NOT make me think about changing my daily routine to stay safe.	Ultimately, the decision about what to do to stay safe is up to me.
This message would make me take the safety risk seriously.	I feel sure that I could figure out the best thing to do for my safety in a situation like this.

3.5 Chain of thought reasoning and Likert scale conversion

To simulate a human response to this questionnaire, the model was then prompted incrementally, as described by Schuller et al., (2024). To promote CoT reasoning, a trait thought to be in line with behavioural responses, the LLM was first prompted to read the provided alert, think out loud, and output the resulting block of commentary.

Following this, the pipeline aimed to convert the LLM’s unfiltered commentary into numerical, Likert-scale ratings. The LLM was prompted with the warning message, the eight questions, and its own commentary from the previous step. It was then instructed to assign Likert ratings for each question, based on the associated reasoning.

Initial testing found this single-shot prompting technique necessary when using the smaller, Llama 3:2.1b. This model achieved a 100% compliance rate with single-shot prompting, compared to 75% with zero-shot prompting. Thus, the pipeline included an exemplar of ratings for the LLM to infer from, after the previous instructions. The commentary

associated with each rating provided enough confidence that responses weren’t giving random values or copying this exemplar. A stylistic prompt template is shown in Figure 1 below.

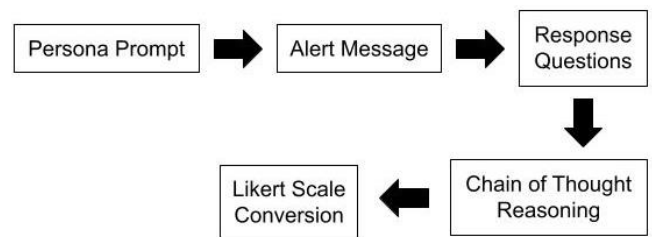


Figure 1: Prompting Structure for LLM pipeline.

3.6 Analysis of results

Carlson and Barbour report their results using an analysis of variance (ANOVA). ANOVA is a statistical method used to compare the means of two or more groups and to determine if their difference is genuine or just a product of noise. It essentially asks: “Are the observed differences in group means larger than what we’d expect by chance?”.

Their research constructed two ANOVA models, the first assessing message efficacy and the second, safety efficacy

with message efficacy as a covariate. This was done to simulate real-world processing, where one would read a message and find it compelling or not, then reflect on how they would act and feel regarding safety.

Carlson and Barbour’s work used a mixed-effects model, treating message replication (the twelve warning messages) as a random factor nested within message strategy (instrumental-only, empathetic, choice-forward). Comparatively, the ANOVA model used in this paper’s analysis was a fixed-effects ANOVA, treating message replication as a fixed factor. This means the model estimates a separate effect for each replication. For the purposes of this report, a fixed-effects model was selected to maintain a practical scope, without extending into more complex hierarchical modelling.

4 RESULTS

4.1 Temperature sensitivity

The effects of temperature were tested on the Llama3:2.1b and Gemma3_27b models. Results showed that as the temperature increased from 0 to 0.3, the standard deviation of scores also increased. Further increases in temperature yielded little change, with standard deviations plateauing for temperatures above 0.3 (Figure 2).

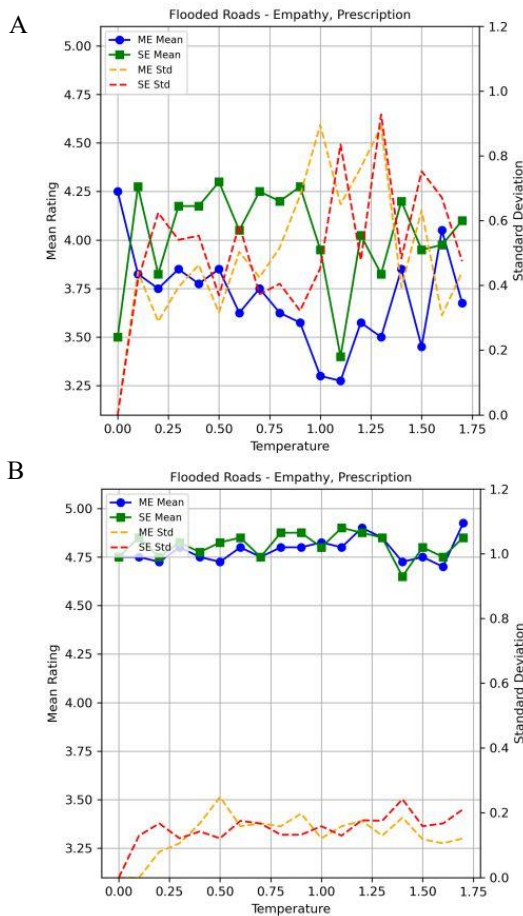


Figure 2: Temperature sensitivity for Llama 3.2.1B (A) and Gemma 3 27B (B) under identical scenarios and message types.

The range of standard deviation for Gemma3_27b was 0.25, while the range for Llama3:2.1b was 0.9. Furthermore, the standard deviation in Llama plateaued at 0.5, whereas Gemma plateaued at 0.1.

Gemma3_27b consistently scored messages in the range of 4-5, insisting they evoked high levels of safety and message efficacy. This was less evident in the Llama model, where the mean SE and ME scores oscillated around 4.

4.2 Persona effects

The most definitive result was observed when persona prompting caused the LLM to act with varying degrees of compliance. The responses show that LLMs prompted to act as a ‘Total sceptic’ provide significantly lower message efficacy scores than those prompted to act as a ‘People-pleaser’ or the baseline, un-persona-prompted model (Figure 3).

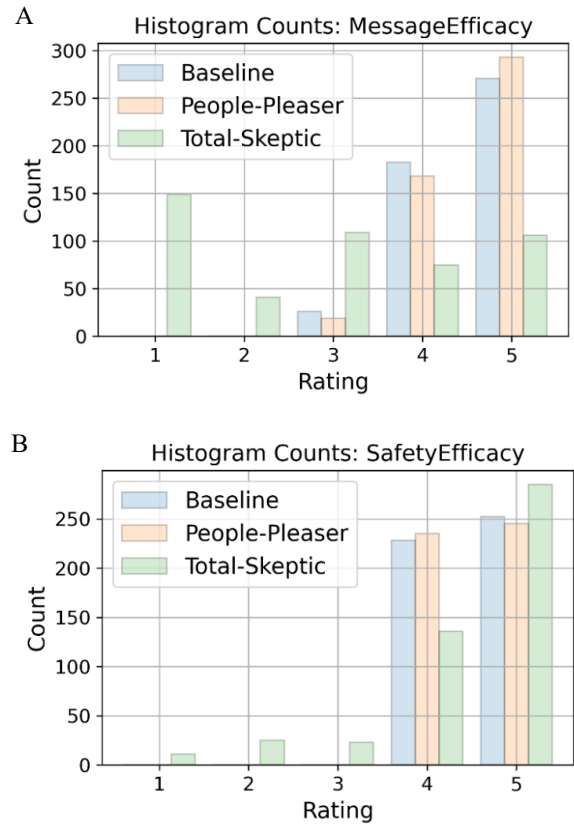


Figure 3: Gemma3_27b histograms for message efficacy (A) and safety efficacy (B).

The total sceptic message efficacy shows a more even distribution in scores, without a skew towards the higher values. This pattern wasn’t reflected in the safety efficacy results, where the total sceptic persona performed similarly to the other two.

The five persona characteristics were visualised using box plots. Two persona pairings did not have any overlap, while the other three pairings did (Figure 4).

twelve different messages produced different message efficacy and safety efficacy ratings.

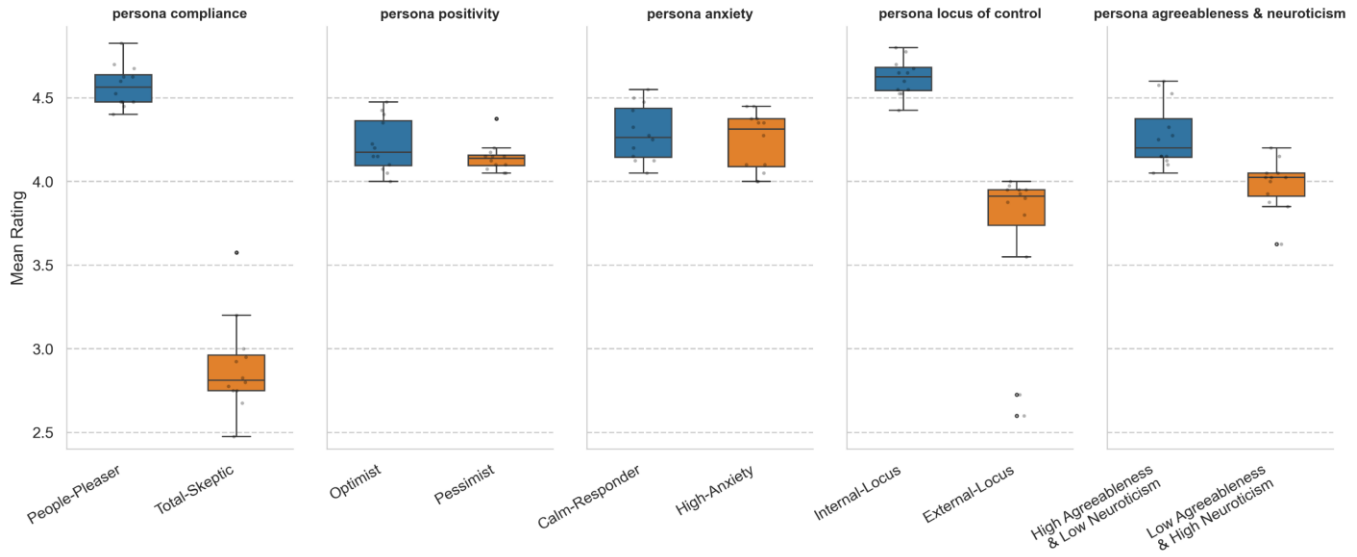


Figure 4: Boxplots for message efficacy of tested persona pairs in Gemma3_27b. Data points represent mean ratings for the twelve different message strategies tested by Carlson and Barbour.

4.3 Validation study replication

ANOVA results from Carlson and Barbour’s study and Gemma3_27b provide a means of validation (Table 4). Using a P-value of 0.05 as the threshold, instances of agreement between methods are highlighted in green and disagreement, red.

This is reinforced with η^2 values of 0.124 and 0.121, respectively, suggesting that around 12% of the witnessed variance is a consequence of the specific message type. Carlson and Barbour did find significance in their responses’ message efficacy with respect to the twelve messages; however, they failed to find significance in the safety efficacy

Table 4: ANOVA for message and safety efficacy, treating message replications as a random factor (Gemma3_27b). Red cells indicate disagreement, and green cells indicate agreement between models.

Variable	Gemma3_27b			Carlson and Barbour		
	F	p	Partial η^2	F	p	Partial η^2
Message Efficacy						
Intercept	13991.018	<0.001	0.98	357.064	0.340	0.997
Replication	3.696	<0.001	0.124	16.516	0.015	0.805
Strategy	2.06	0.129	0.014	1.310	0.266	0.019
Direction	2.761	0.098	0.01	5.223	0.023	0.019
Safety Efficacy						
Intercept	138.577	<0.001	0.329	5,557.323	0.005	1.000
Replication	3.535	<0.001	0.121	0.191	0.684	0.044
Strategy	0.934	0.394	0.007	3.537	0.018	0.331
Direction	0.875	0.350	0.003	6.396	0.014	0.094
Message Efficacy	1.436	0.232	0.005	7.229	<0.001	0.594
Interaction of message strategy and message direction	0.153	0.858	0.001	0.603	0.698	0.015
Interaction of message strategy and message efficacy	0.767	0.465	0.005	2.065	<0.001	0.333
Interaction of message direction and message efficacy	1.196	0.275	0.004	2.147	0.023	0.100

In the case of Gemma3_27b, the intercept was found to be statistically significant with p-values < 0.05 in each ME and SE scenarios. This means that the average Likert ratings across all responses were high enough to be statistically significant. This is validated by the mean scores shown by Gemma3_27b in Figure 1. Carlson and Barbour found similar significance in safety efficacy but did not observe the same significance in message efficacy.

Message replication was also found to have statistical significance in Gemma3_27b’s analysis, suggesting that the

responses and produced markedly different η^2 values.

The significant results that Carlson and Barbour found regarding safety efficacy were not replicated. Their research found message strategy (empathetic, choice-forward, or instrument-only), direction (prescriptive or notification), message efficacy, and the interaction of message strategy and message efficacy all to have influence over safety efficacy scores. Comparatively, Gemma3_27b found no significance in treating safety efficacy as a random factor in anything other than intercept and message replication.

5 DISCUSSION

5.1 Interpretation of Findings

It was hypothesised that increasing temperature would linearly increase the standard deviation of Likert responses, reflecting greater variability in participant responses. This hypothesis was not supported, as the standard deviation plateaued at a temperature of 0.3. While temperature remains important to the function of LLMs, its effect on response variability was less than initially thought. Results did, however, provide a useful temperature range.

Model size was found to have a greater influence on results. It could be suggested that the larger variability in the Llama3:2.1b responses more accurately recreated human response variability. However, analysis of the commentary and scores for the smaller model showed that its reasoning did not align with the scores it returned. Instead, the smaller model could have returned a random answer just to comply. This is where the larger standard deviation came from, as making educated guesses would still return natural variability. This method of introducing variability is unsatisfactory as it does not reflect the same factors that create differences in human populations. A more effective approach would be to replicate human diversity through persona prompting.

While the total sceptic persona had a significant effect on message efficacy, it showed little effect on safety efficacy. This suggests that while the sceptic prompt had a strong effect on reducing the LLM's willingness to comply, it did not directly diminish its concern for safety. This is a positive finding with regard to RQ2, as it demonstrates that some persona prompts can sometimes steer model outputs in a desired direction. This shows that prompt engineering can provide controlled and deliberate manipulation of LLM responses.

The effect of persona prompting on safety efficacy was less conclusive, showing less variance between persona-pairs. This could stem from message efficacy being more subjective, as there are multiple parts of a message that could affect its compliance. Safety efficacy is a more binary subject, as a message will either make you act safely or not. Furthermore, safety behaviours are encoded into LLMs with alignment training and system prompts, which might limit the influence of persona prompts. This suggests that while personas can perceive effectiveness, they may have less influence on safety interpretation.

ANOVA results were equally concerning for LLM safety efficacy replication. None of the significant findings found by Carlson and Barbour could be replicated, namely, a higher message efficacy didn't correlate to a higher safety efficacy; a result that should logically be present. This could be because humans associate message efficacy with the risk of the emergency, while LLMs separate the two, and rank the messages based more on linguistic clarity. Alternatively, the LLM may be unable to replicate human results, scoring either four or five, irrespective of the persona prompts. Another rationale is that the CoT reasoning in the pipeline may cause the LLM to evaluate safety and message efficacy separately, rather than as a human would consider them together. These findings align with Bisbee et al., (2024), who similarly found that, while LLMs can approximate overall human response patterns, they often distort important nuances.

5.2 Further research

Beyond the scope of this study, different models of varied parameter size could be compared to see LLM progression overtime. This study only assessed four models but compared Gemma3_27b and Gemma3_12b, finding that the larger model provided results with greater certainty. Models come much larger and computationally more efficient than these, providing a future research avenue. Larger models come with the costs of greater computational power and increased run time. Optimising model size to create accurate human results without the diminishing returns associated with larger models would also be a valuable direction for future research.

Brief research was conducted into the difference in models of a similar size, namely the Deepseekr1_14b and Gemma3_12b models. No significant differences between results were found, despite the different training data and upstream commands each model has. This is a path for further exploration, with the availability of different models only expanding. Furthermore, specialised models such as Centaur (Binz et al., 2025), specifically trained on psychological data and experiments, could be examined in this context.

Further research could test this pipeline in other emergency contexts, where response times and thought processes differ. For the example of an earthquake, human behaviour becomes more instinctual and requires immediate action. These sudden events are an area that the theory of reasoned action could develop within. As this experiment allowed analytical decision-making that reduced reasoned-action bias, future applications could instead focus on instinct-driven contexts like fire or earthquake evacuation.

The most influential variable in the pipeline was personas, with the prompting of personas providing notably different outputs. A variety of personas were examined; however, they were one-dimensional and intentionally exaggerated, like the total sceptic vs people pleaser, to better observe the impact of extreme behavioural cues. These results showed that persona prompting can manipulate LLM output, providing a foundation for further research. With different personality models, such as the Five Factor model (McCrae & John, 1992), there is room for exploration on which personality traits are the most impactful. Such models could also assist in creating human-like personalities with multi-dimensional traits. The use of personas to create sample populations is another line of investigation.

6 CONCLUSION

This study demonstrates the potential for LLMs to act as scalable human participants in studies for emergency alert systems. Persona prompting, for one-dimensional persona traits, proved the most influential factor for replicating human scores, though some personas were more impactful than others. While the results suggest that LLMs could provide an efficient and effective alternative to human subjects, there are still limitations with a lack of lived emotional experience, model size, and reliance on exaggerated personas. Future research should explore different personality models and models trained on psychological data across varied emergency contexts to improve the applicability of LLMs as simulators of human participants. If proven to be a robust method, this will provide an ethical alternative to traditional human surveys.

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